The Learning-Curve Treadmill: Wright's Law, Capacity Reinvestment, and Downturn Risk in South Korea and Taiwan

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Abstract

South Korea and Taiwan hold the two most concentrated semiconductor positions in the world economy. In October 2026 semiconductors were 49.0 percent of Korean exports (Reconstructed, B), and Samsung Electronics with SK hynix were 42.2 percent of KOSPI capitalisation in May 2026 (Observed, B). TSMC was about 43 percent of the TAIEX in November 2025 (Observed, B). This paper asks whether reinvesting boom surpluses into capacity on steep learning curves amplifies the damage when demand pauses. Three computed results frame the answer. First, the DRAM learning rate estimated in KR-WP-2026-05 at 38.6 percent reproduces exactly on the parent data and steepens across subperiods, from 35.1 percent in 1971–1990 to 47.4 percent in 1991–2007 (Reconstructed, B), while TSMC revenue per wafer rises with cumulative output, an apparent learning rate of −34.0 percent (Reconstructed, C): the treadmill is a memory phenomenon, not a foundry phenomenon, in the data held. Second, in the four downturn episodes since 2000, Korean chip exports fell 19 to 28 percent and Samsung group operating profit fell 39 to 85 percent, while TSMC revenue fell at most 24 percent and rose 4.5 percent in 2019. Third, at the fitted DRAM curve, bit growth of 20 percent per year implies a price decline of 12.0 percent per year, so volume must grow 13.7 percent per year to hold revenue flat; a representative new fab breaks even only if prices fall less than 37.0 percent at full utilisation and less than 29.3 percent at 70 percent utilisation (Modelled, C), against observed episode price falls of 43 to 59 percent. Applying the 2022–23 episode rates to the 2025 base puts Korean semiconductor exports US$41.1 billion lower and SK hynix at an operating loss (Modelled, B/C). The thesis holds for commodity memory and fails, so far, for contracted HBM and leading-edge foundry output. The loss order in a crash runs from memory-firm equity to Korean policy lenders; TSMC enters with operating cash flow above capital expenditure and no policy-loan channel of comparable size.

Keywords: Wright's Law; learning curves; semiconductors; concentration risk; South Korea; Taiwan; industrial policy.
JEL: F14, L63, O33, G32.

Non-technical summary. Korea and Taiwan earn a large share of their export income from chips whose prices fall by a predictable amount every time total production doubles. That price fall is good for buyers and hard for sellers: to earn the same revenue, sellers must ship much more each year. Both economies are now spending boom profits on new factories. This paper measures the price falls, the concentration, and the past downturns, then does the arithmetic. The arithmetic says the new memory factories now being built stop covering their fixed costs if prices fall by roughly

30 to 37 percent, and prices fell by more than that in each of the last four downturns. The same arithmetic does not bind TSMC, whose price per wafer has risen, not fallen. If demand pauses, the first losses land on Korean memory companies and on the Korean state banks that lend to them at below-market rates. This is a retrospective analysis; there was no pre-registered plan.

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1 Introduction

The question is narrow. When an economy's exports, listed-market value, and industrial base concentrate in products on steep learning curves, does reinvesting boom surpluses into still more capacity amplify, rather than cushion, the damage when demand pauses, because prices keep falling along the curve while the new capacity's fixed costs do not?

The timing is set by one month of trade data. In October 2026 Korean exports were US$115.28 billion, semiconductor exports were US$56.44 billion, and the trade surplus was US$42.50 billion (Observed, B; Table 1). The surplus is the reinvestment fund. Where it is going is also on record: TSMC's 2026 capital budget is US$60–64 billion (Observed, A), SK hynix's board approved KRW 54.3 trillion for two fabs in August 2026 (Observed, B), and the Korean state has raised its semiconductor support package to KRW 33 trillion (Observed, B).

The paper computes five blocks and reports each as a table or a figure: learning curves (Section 6.1), concentration (Section 6.2), downturn episodes (Section 6.3), treadmill and break-even arithmetic (Section 6.4), scenario estimates and the financing map (Sections 6.5 and 6.6). Prose is limited to the question, the method, and the result.

Preview of results. (1) The memory treadmill is real and steepening in the fitted data. (2) Concentration is at record measured levels in both countries. (3) Downturn losses scale with commodity-memory exposure, not with concentration as such: the pure memory producer takes the largest loss in every episode, the diversified Korean group takes the second largest, and the foundry takes the smallest. (4) Break-even price falls for new memory capacity sit below the price fall in every observed episode since 2000. (5) The financing map is asymmetric: Korea has a standing policy-loan channel into the exposed firms; Taiwan's exposure runs through the equity market and a 6.38 percent state fund shareholding in TSMC (Observed, B).

Falsifiability. The thesis is rejected, in whole or in part, if any of the following holds in data that become available: HBM and other contracted output durably prices off the learning curve at a share of Korean chip exports large enough to move the aggregate (the paper tests the current state of that claim in Section 7); the two-firm KOSPI weight and the semiconductor export share fall back below their 2023 levels while capacity is still being added; or a memory price fall of 40 percent or more occurs without a fall in Korean chip exports, a policy-loan drawdown, or a broad-index earnings decline. Section 7 scores the present evidence against each test.

The rest of the paper proceeds as follows. Section 2 sets out how the two industries sell and finance output. Section 3 places the estimates in the literature. Section 4 describes the data and its gaps. Section 5 states the computations. Section 6 reports the five blocks. Section 7 covers limitations, sensitivity, and the falsification scorecard. Section 8 concludes. The data appendix defines every series.

2 Institutional background

Memory is sold in two ways. Commodity DRAM and NAND trade on contract and spot prices that reset quarterly or faster, and the price record in Section 6.3 shows falls of 43 to 59 percent inside single downturns. HBM is sold on annual contracts negotiated with a small number of accelerator buyers; reporting on the 2026 cycle records long-term agreements that share downside between buyer and seller (The Economist 2026). Foundry output is sold wafer by wafer under capacity reservations, and TSMC's blended revenue per wafer has risen since 2018 (Table 3), because the mix shifts toward smaller nodes faster than per-node prices fall.

The two national positions differ in structure. Korea's position is two firms making a commod-

ity. Taiwan's position is one firm selling a service with no close substitute at the leading edge. That structural difference, not the headline concentration ratios, is what the episode evidence in Section 6.3 prices.

Capacity is lumpy and late. SK hynix's August 2026 board approval carries first cleanrooms in December 2028 (Cheongju M17) and June 2029 (Yongin Y2) (Observed, B). Samsung suspended its Pyeongtaek P5 line in 2024 and restarted it for a 2028 opening (Observed, B). Capacity decided at boom prices therefore produces at post-boom prices in the normal case, which is the timing assumption the break-even arithmetic in Section 6.4 prices.

Table 1: Announced capacity, support, and the October 2026 trade outturn

Item	Amount	When
TSMC capital expenditure, actual	US$40.9bn	2025
TSMC capital budget, guided	US$60–64bn	2026
TSMC total planned US investment	US$165bn	Mar 2025
SK hynix board-approved fabs (Y2, M17)	KRW 54.3tn	Aug 2026
SK hynix Yongin cluster framework	KRW 600tn	to 2033
Samsung domestic investment plan	KRW 450tn / 5 yrs	Nov 2025
Korea support package	KRW 33tn	Apr 2025
Korea exports, semiconductors	US$56.44bn	Oct 2026
Korea exports, total	US$115.28bn	Oct 2026
Korea trade surplus	US$42.50bn	Oct 2026

Notes: Amounts as announced; frameworks are authorisations, not spending. Source: data/announcements.csv. TSMC items from company results and releases (A); Korean items from company and government announcements via press reporting (B).

3 Related literature

Wright (Wright 1936) fitted unit cost against cumulative output for airframes. The form used here, $\ln C = a + b \ln Q$, is the standard estimating equation in the survey literature (Sahal 1981; Nagy et al. 2013). KR-WP-2026-05 (Kilambi Research 2026) estimated the DRAM rate at 38.6 percent on 1971–2007 data and reported that cumulative output and calendar time are near-collinear, so bivariate time-augmented fits are not read structurally; this paper keeps the bivariate Wright form for the same reason and reads the foundry fit as a mix diagnostic, not as a learning estimate. The memory cycle literature documents the same price pattern at higher frequency: Gartner's 2019 review records a 47.4 percent DRAM average selling price decline in 2019 on oversupply that began in late 2018 (Gartner 2020). AMRO's 2026 Korea consultation annex records the export-share series, the two-firm global DRAM share of about three-quarters, and a 170 percent year-on-year DRAM contract price rise in Q3 2025 (ASEAN+3 Macroeconomic Research Office 2026): the curve binds in both directions. The concentration framing follows the export-concentration and terms-of-trade volatility literature; no new method is claimed.

4 Data

Eight data files, all in the replication package, carry every number in the paper. Table 2 summarises the estimation samples. Definitions and per-series sources are in Appendix A.

Table 2: Summary statistics: estimation and measurement samples

Series	Unit	n	Span	Grade
DRAM price and cumulative output	2005 US$/kbit; Mbits	37	1971–2007	B
TSMC revenue per wafer	NT$ thousand per wafer	8	2018–2025	C
Firm panel (3 firms)	KRW tn; US$ bn; percent	27	2017–2025	B
Korean exports, semiconductors	US$ bn	9	2017–2025	B
Taiwanese exports and TSMC revenue	US$ bn	10	2015–2024	B
Market-cap concentration points	percent	8	2025–2026	B/C
Downturn episodes	percent changes	4	2000–2023	B/C
Announcements	currency amounts	8	2024–2026	A/B

Notes: n counts annual observations per series (firm panel: 9 years times 3 firms). Grades use the desk vocabulary: A official or primary, B reputable secondary, C provisional. Source: data/ directory of the replication package.

Named gaps, stated before the results because they bind what can be computed. (1) NAND: no public annual series pairs NAND price per bit with cumulative bit output over a long span in held data, so no NAND Wright fit is reported; NAND enters through episode price falls only. (2) HBM: HBM prices are contract prices and are not published as a series; no HBM fit is reported. HBM enters through contract-structure evidence and the falsification scorecard. (3) Korean manufacturing capex by firm is not published on a consistent basis, so the capex concentration block reports investment intensity (capex relative to revenue and to exports) rather than a national capex share. (4) The PCDB DRAM series ends in 2007; the curve is therefore estimated on the parent sample and the post-2007 evidence enters through episode ASP changes, not through a refit. (5) A single audited aggregate for US hyperscaler and datacenter debt is not in held data; the financing map reports the supply side in amounts and the demand side as a named gap.

5 Methodology

Learning curves. For each product with a price and cumulative-output series we estimate

$$\ln P_t = a + b \ln Q_t + u_t, \tag{1}$$

by OLS with heteroskedasticity-consistent standard errors. The learning rate is $LR = 1 - 2^b$: the proportional price fall when cumulative output doubles. Confidence bands in Figure 1 are the OLS prediction bands from the same covariance matrix.

Treadmill. Revenue is $R = P(Q) \times V$ with $P(Q) = P_0 Q^b$. At annual cumulative-output growth g, the implied annual price change is $(1+g)^b - 1$, and the volume growth that holds revenue constant is $1/(1 + \Delta P) - 1$.

Break-even. A representative new fab is indexed to capex 100 and revenue 100 at current prices and full utilisation. Variable costs are 45 percent of revenue (Assumed, C). Fixed costs per year (depreciation plus fixed operating cost) are varied at 12, 18, and 24 percent of capex. The break-even price fall at utilisation u solves $(1 - x) \cdot 100u - 45u - F = 0$. The base case ($F = 18$) is set below SK hynix's 2025 company operating margin of 49 percent (Observed, B), because a single fab carries no HBM packaging margin.

Scenarios. Episode percentage changes from Table 5 are applied to 2025 levels from the firm panel and the export series. Index earnings impacts are arithmetic: index weight times the blended

firm earnings change. Every scenario number is Modelled and its assumption is printed in the table note.

6 Results

6.1 Learning curves

Table 3: Estimated equation $\ln P = a + b \ln Q$; dependent variable is log price

Sample	n	b (SE)	Learning rate	95% CI	R^2
DRAM, 1971–2007	37	−0.704 (0.017)	38.6%	[37.1, 40.0]	0.984
DRAM, 1971–1990	20	−0.623 (0.022)	35.1%	[33.1, 36.9]	0.986
DRAM, 1991–2007	17	−0.928 (0.039)	47.4%	[44.6, 50.2]	0.972
TSMC revenue per wafer, 2018–2025	8	0.422 (0.094)	−34.0%	n.a.	0.773
NAND	Not estimated: named gap 1, Section 4				
HBM	Not estimated: named gap 2, Section 4				

Notes: Standard errors in parentheses, heteroskedasticity-consistent. The TSMC slope is positive: blended revenue per wafer rose with cumulative shipments on a partial cumulative sum from 2018, because node mix dominates per-node price declines. It is a mix diagnostic, not a learning estimate. n.a.: the interval is not reported for an 8-observation diagnostic fit. Source: results/learning_fits.csv; data/dram_wright.csv; data/tsmc_wafers.csv.

Result 1. The DRAM curve is steep and steepens: the 1991–2007 learning rate of 47.4 percent (Reconstructed, B) exceeds the 1971–1990 rate of 35.1 percent by 12.3 percentage points, and the two confidence intervals do not overlap. The foundry proxy has the opposite sign. Any aggregate "semiconductor learning rate" for these two economies is therefore a weighted average of a steep negative (memory) and a positive (foundry mix) slope, and the weight is the commodity share of exports. Korea's weight on the negative slope is the larger one: memory is about two-thirds of Korean semiconductor exports (Observed, B; ASEAN+3 Macroeconomic Research Office 2026).

6.2 Concentration

Result 2. Concentration rose on every measured margin. The Korean semiconductor export share moved from 20.9 percent in 2018 to 15.6 percent in 2023 and 24.4 percent in 2025 (Reconstructed, B), and reached 49.0 percent in the single month of October 2026 (Observed, B). The two-firm KOSPI weight rose from 22.6 percent in January 2025 to 42.2 percent in May 2026 (Observed, B). TSMC revenue rose from 9.25 percent of Taiwanese exports in 2015 to 19.0 percent in 2024 (Reconstructed, B). The 2023 observation matters for interpretation: the export share fell 5.3 percentage points in one downturn year, which is the transmission channel measured directly. TSMC's capital expenditure reached 33.4 percent of revenue in 2025, US$40.9 billion on US$122.4 billion of revenue (Reconstructed, B; computed in results/firm_panel_computed.csv), and the 2026 budget of US$60–64 billion raises that share at any revenue growth rate below the budget's own growth rate.

6.3 Downturn episodes

Result 3. Losses order by product, not by country. Across the four episodes, SK hynix (pure memory) records the largest operating swing in every case, Samsung (diversified, memory-heavy)

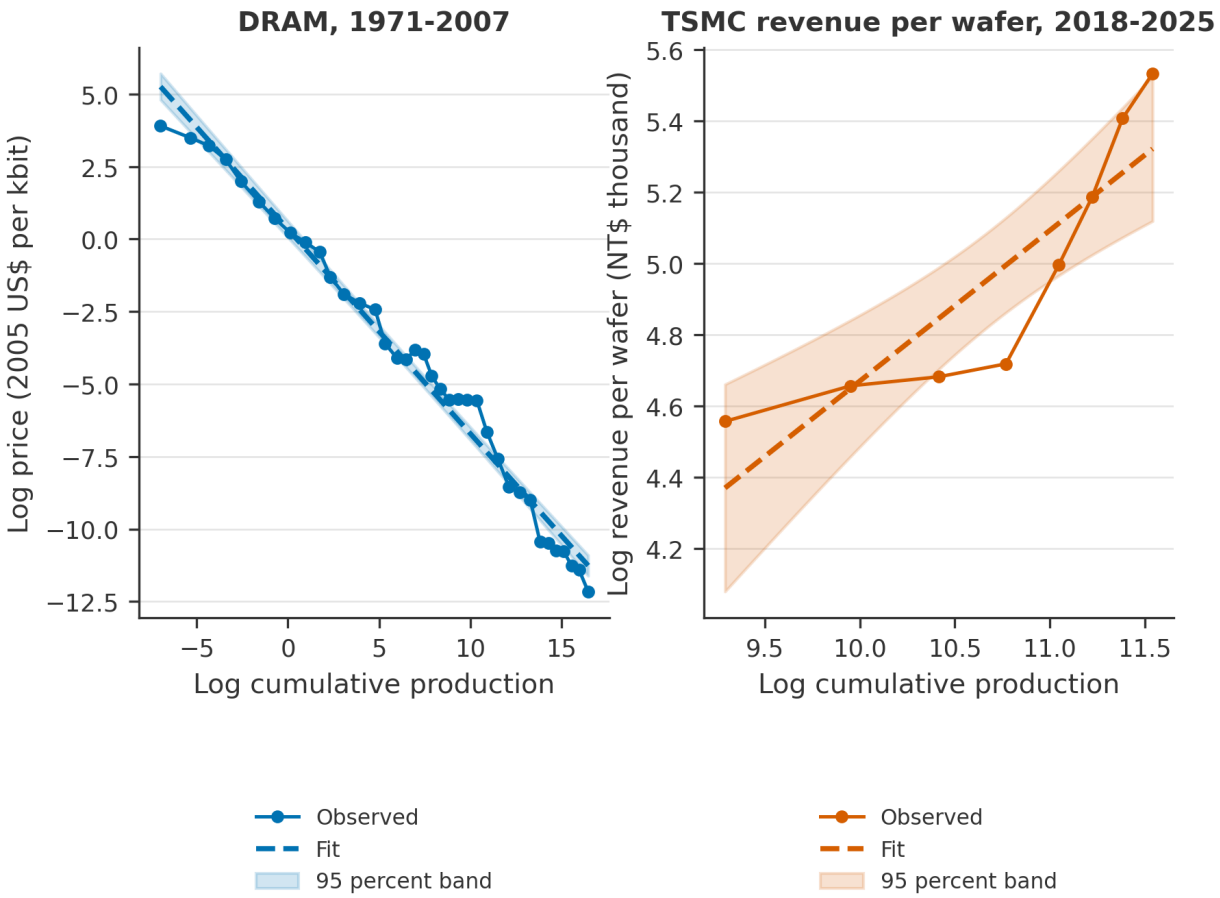


Figure 1: Fitted curves with 95 percent bands: DRAM and TSMC revenue per wafer
Source: results of Table 3. Note: Left panel, DRAM price falls along the curve. Right panel, TSMC revenue per wafer rises with cumulative shipments from a 2018 base (partial cumulative), so the fitted slope has the opposite sign.

Table 4: Dated market-capitalisation concentration points

Date	Market	Firms	Share (%)
Jan 2025	KOSPI	Samsung Electronics + SK hynix	22.6
Mar 2026	KOSPI	Samsung Electronics + SK hynix	40.6
Apr 2026	KOSPI	Samsung Electronics + SK hynix	41.0
May 2026	KOSPI	Samsung Electronics + SK hynix	42.2
Jun 2026	KOSPI, monthly avg	Samsung Electronics + SK hynix	55.7
Nov 2025	TAIEX	TSMC	43.0
May 2026	TAIEX	TSMC	42.0
Jun 2026	TWSE total	TSMC	>40.0

Notes: KOSPI points from Korea Exchange data via Seoul Economic Daily and AJU Press (B); the June 2026 monthly-average point is Korea Financial Investment Association data via secondary reporting (C) and is not on the same measurement basis as the dated closes. TAIEX points are index weights compiled by Bloomberg via The Edge (B). Source: data/concentration_points.csv.



Figure 2: Export concentration: Korea and Taiwan
Source: data/korea_exports.csv; data/taiwan_tsmc_share.csv (B). Note: Top panel, semiconductors as a share of Korean exports. Bottom panel, TSMC revenue as a share of Taiwanese exports (Reconstructed: company revenue divided by national goods exports; a scale ratio, not a customs category).

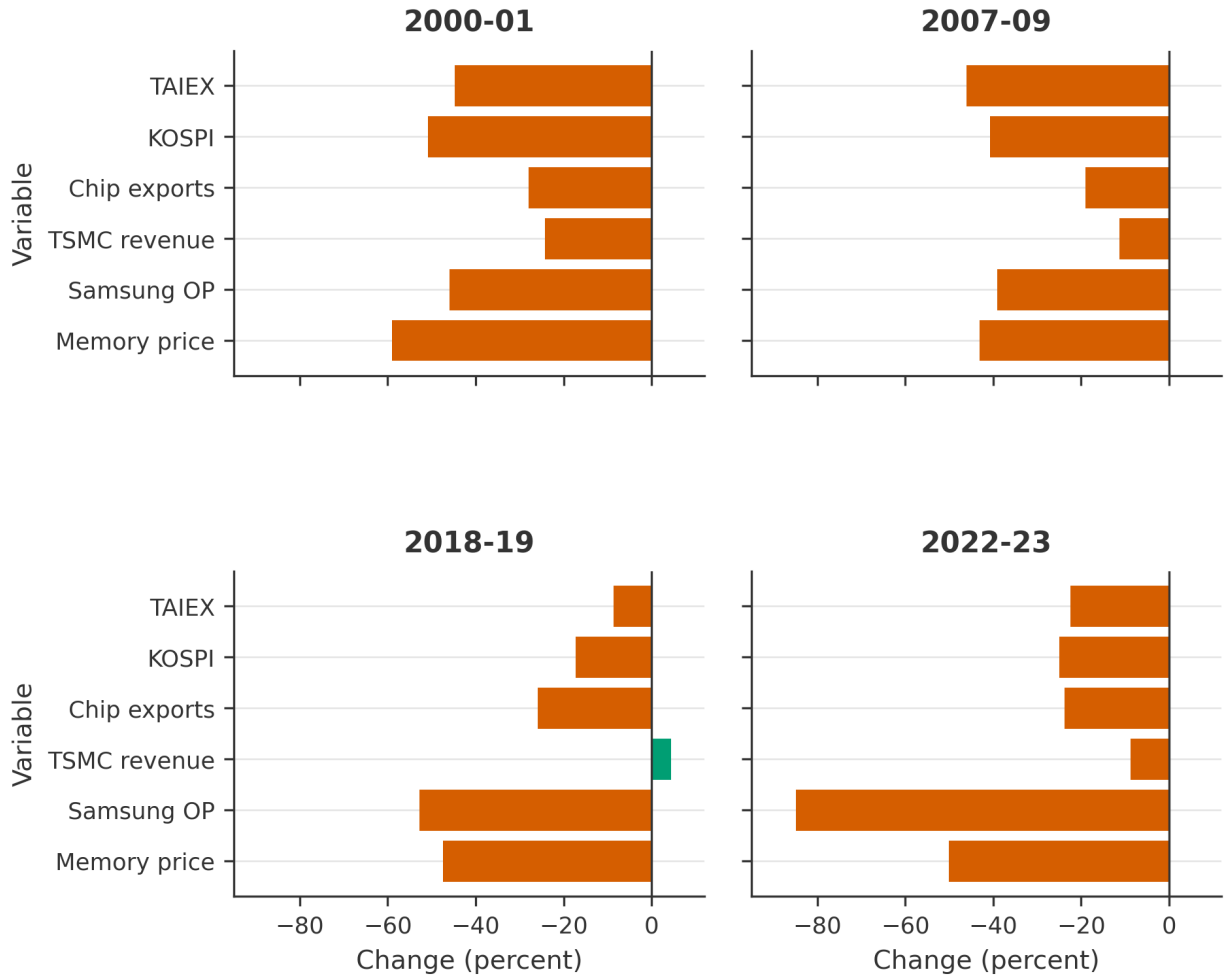


Figure 3: Event-study panels: four downturn episodes on shared axes
Source: data/episodes.csv. Note: Each panel aligns one episode on the same variables and the same x-axis range (H5). SK hynix is excluded from the panels because its profit-to-loss swings (−87 to −214 percent) would compress every other bar; its values are in Table 5.

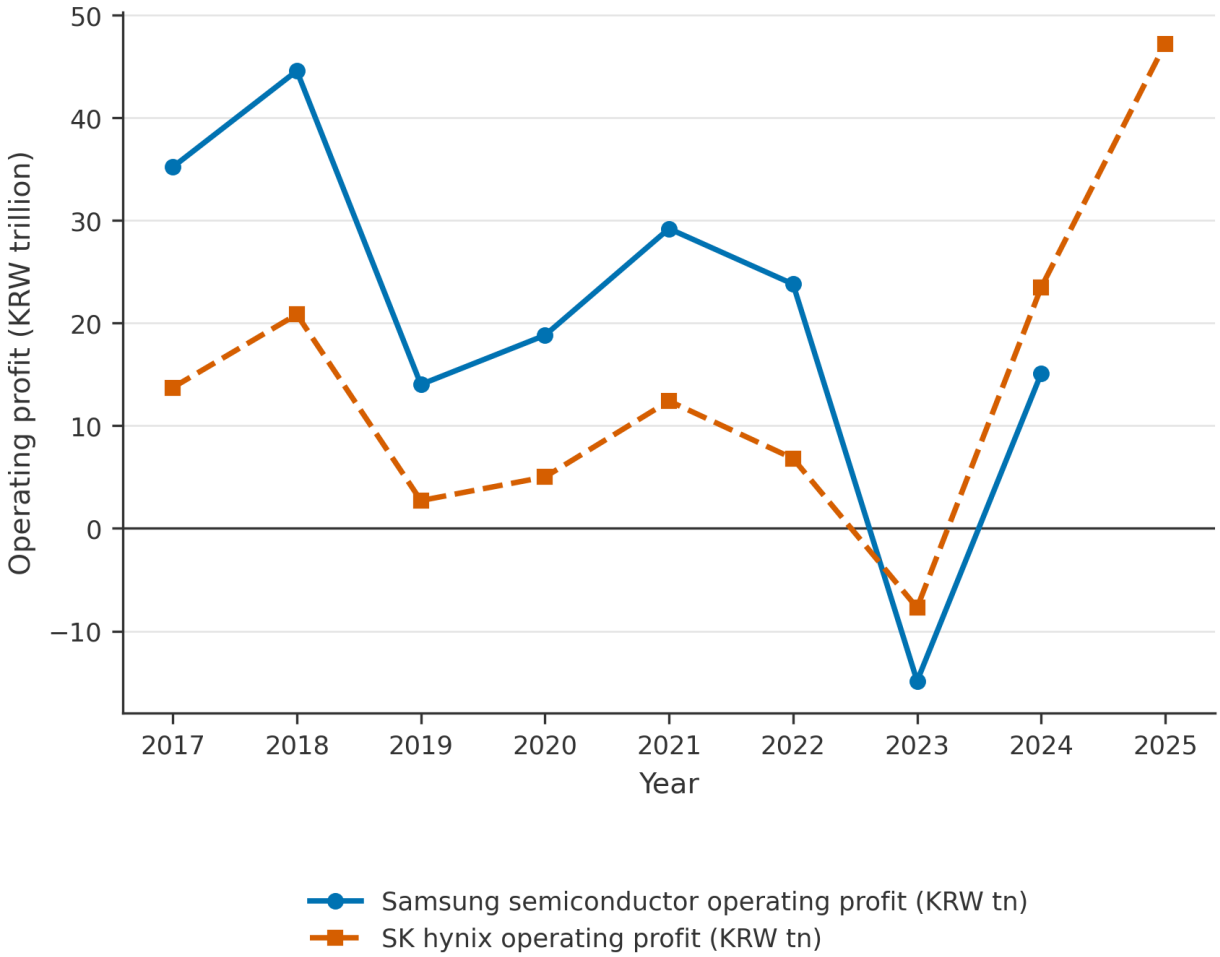


Figure 4: Operating profit: Samsung semiconductor division and SK hynix, 2017–2025 Source: data/firm_panel.csv (B). Note: Samsung semiconductor moves from KRW 23.8 trillion profit in 2022 to a KRW 14.9 trillion loss in 2023, a swing of KRW 38.7 trillion (Reconstructed, B). SK hynix moves from KRW 6.81 trillion to a loss of KRW 7.73 trillion over the same years, and to a record KRW 47.2 trillion profit in 2025.

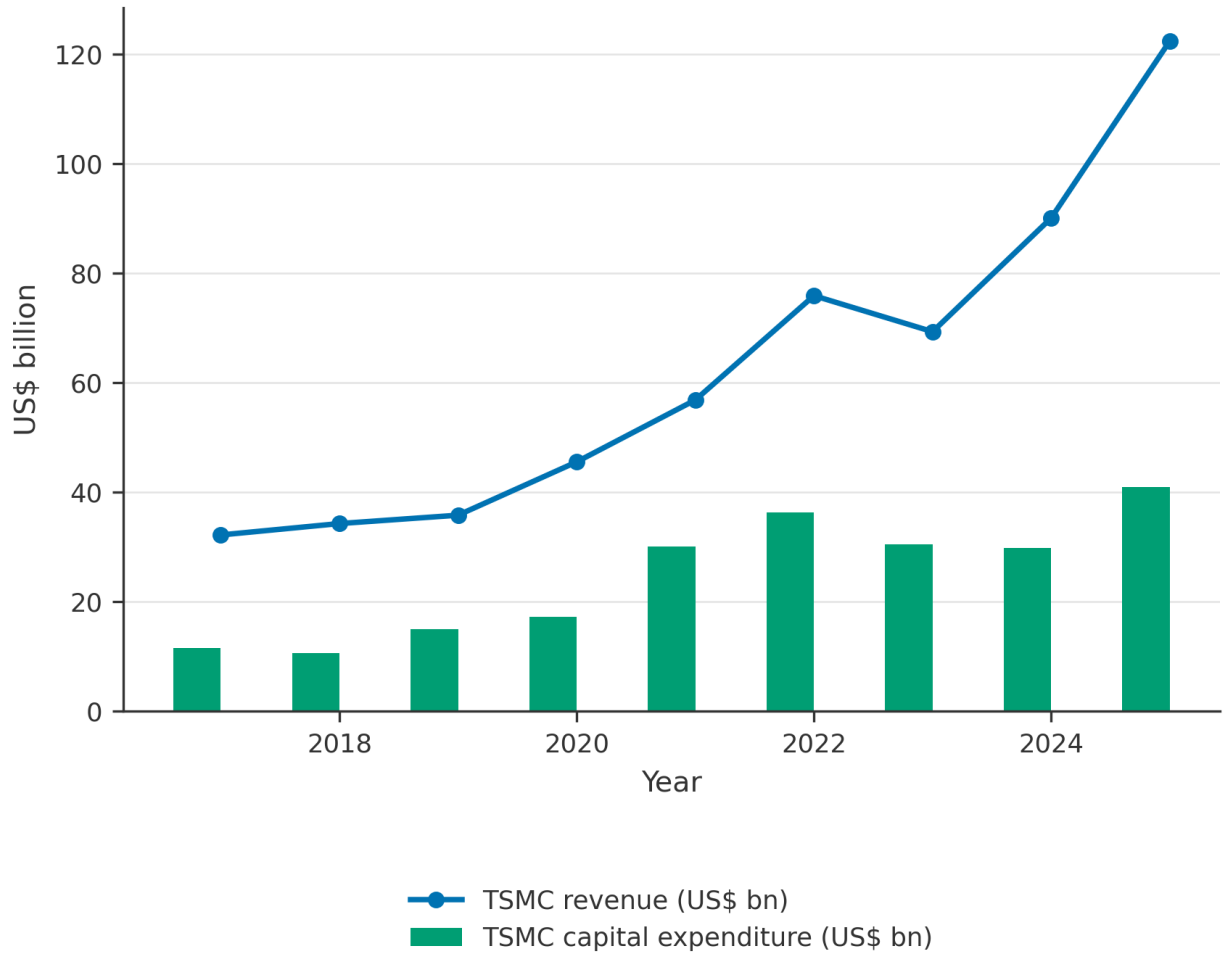


Figure 5: TSMC revenue and capital expenditure, 2017–2025
Source: data/firm_panel.csv (B). Note: Revenue fell in one year of the nine (2023, −8.7 percent in US dollars) while capital expenditure rose through the downturn, from US$30.4 billion in 2023 to US$40.9 billion in 2025.

Table 5: Downturn episodes: standardised comparison (percent changes)

Variable	2000–01	2007–09	2018–19	2022–23
Memory price	−59.0	−43.0	−47.4	−50.0
Global semiconductor sales	−32.0	−9.0	−11.9	−8.2
Samsung group operating profit	−46.0	−39.0	−52.8	−84.9
SK hynix operating profit	−112.0	−118.0	−87.0	−213.5
TSMC revenue	−24.3	−11.3	4.5	−8.7
Korean chip exports	−28.0	−19.0	−25.9	−23.7
KOSPI (calendar year)	−50.9	−40.7	−17.3	−24.9
TAIEX (calendar year)	−44.7	−46.0	−8.6	−22.4

Notes: SK hynix values below −100 denote a swing from profit to loss, scaled on base-year profit. Memory price 2018–19 is the DRAM ASP change (Gartner); 2022–23 is NAND (about −50 percent year on year, secondary reporting). Early-episode firm values are approximate (B/C). Calendar-year index changes are not peak-to-trough drawdowns. Source: data/episodes.csv.

the second largest, and TSMC (foundry) the smallest revenue change in every case, including a rise in 2019. Korean chip exports fall in a narrow band, 19 to 28 percent, in all four episodes. Broad indices fall in all eight country-episodes, but the falls do not scale with the chip-export falls: the two largest index falls (2000–01, 2007–09) coincide with global equity bear markets, so the index row is evidence of co-movement, not of chip-specific transmission. The chip-specific transmission is in the export and firm rows. Export falls are narrower than price falls in every episode, 19 to 28 percent against 43 to 59 percent, because bit volumes rose through the downturns; the gap between the two is the volume offset the treadmill arithmetic of Section 6.4 has to produce.

6.4 The treadmill and break-even capacity

Table 6: Treadmill arithmetic at the fitted DRAM slope ($b = -0.704$)

Bit growth (%/yr)	Price change (%/yr)	Volume growth for flat revenue (%)	Revenue change (%)
10	−6.5	6.9	2.9
15	−9.4	10.3	4.2
20	−12.0	13.7	5.5
25	−14.5	17.0	6.8
30	−16.9	20.3	8.1

Notes: Modelled from the Table 3 slope. Revenue change assumes volume grows at the bit growth rate. Source: results/treadmill.csv.

Result 4. The treadmill binds at observed magnitudes. Holding revenue flat on the fitted curve requires volume growth of 13.7 percent per year at 20 percent bit growth, rising to 27.5 percent at the late-sample learning rate and 30 percent output growth (Modelled); at that corner of Table 7 the implied price decline is 21.6 percent per year. New capacity breaks even only inside a 21 to 43 percent price fall across the whole sensitivity grid, and episode price falls were 43 to 59 percent. The conclusion does not rest on the base case: it holds in every grid cell at or below the observed episode range, and the one cell at the boundary (43.0 percent) is the cell with the lowest fixed costs at full utilisation, the state a downturn removes first.

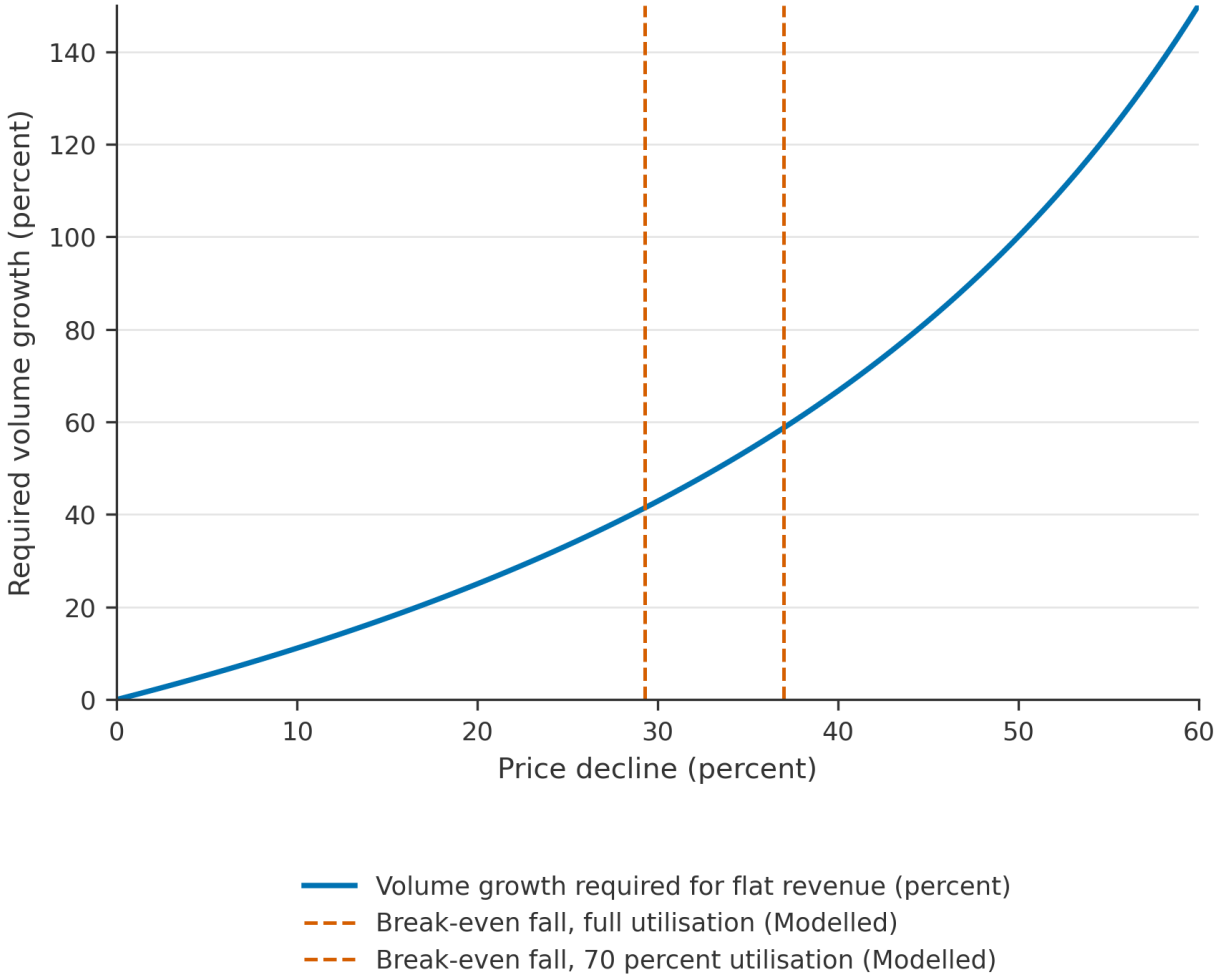


Figure 6: Required volume growth for flat revenue, and break-even markers
Source: results/treadmill.csv; results/breakeven.csv. Note: The dashed lines mark the base-case break-even falls of Table 8 at 100 and 70 percent utilisation. A 47 percent price fall, the 2019 observation, requires volume growth of 89 percent to hold revenue flat.

Table 7: Sensitivity: implied annual price decline and required volume growth, by learning rate

	Cumulative output growth (%/yr)			
Learning rate (%)	15	20	25	30
20.0	−4.4 / 4.6	−5.7 / 6.0	−6.9 / 7.4	−8.1 / 8.8
30.0	−6.9 / 7.5	−9.0 / 9.8	−10.8 / 12.2	−12.6 / 14.5
38.6	−9.4 / 10.3	−12.0 / 13.7	−14.5 / 17.0	−16.9 / 20.3
47.4	−12.2 / 13.8	−15.5 / 18.4	−18.7 / 23.0	−21.6 / 27.5

Notes: Each cell is price decline / required volume growth, percent per year (Modelled). The 38.6 row is the full-sample fit; the 47.4 row is the 1991–2007 fit. Source: results/sensitivity_lr.csv.

Table 8: Break-even price fall for a representative new fab (percent)

Fixed cost (% of capex/yr)	Utilisation 100%	85%	70%
12	43.0	40.9	37.9
18	37.0	33.8	29.3
24	31.0	26.8	20.7

Notes: Modelled (C). Variable costs 45 percent of revenue (Assumed, C); revenue indexed to 100 at current prices. The break-even fall is the price decline at which contribution equals fixed cost. Source: results/breakeven_grid.csv.

6.5 Scenario estimates

Result 5. A repeat of either modern episode removes about a quarter of Korean semiconductor export revenue (US$41 to 45 billion, Modelled, B) and, in the 2022–23 calibration, returns Samsung group operating profit to its actual 2023 level and SK hynix to a large loss. The Taiwan rows move by an order of magnitude less, and in the 2018–19 calibration TSMC revenue rises. The asymmetry is the paper's central estimate: identical global demand shocks produce a Korean macro event and a Taiwanese firm-level event, because the products sit on different curves.

6.6 Who funds the capacity

Result 6. The loss order follows the table. First, memory-firm equity: the 2023 episode destroyed KRW 38.7 trillion of Samsung semiconductor operating profit in one year without any public support being drawn (Reconstructed, B). Second, Korean policy lenders: the KRW 20 trillion low-interest loan programme (Observed, B) puts the state bank inside the capital structure before a downturn, at rates discounted by up to 120 basis points for large fab operators (Observed, B), and the 2001 Hynix restructuring is the observed precedent for creditor-led absorption when equity is exhausted. Third, the Korean taxpayer, through the fiscal parts of the KRW 33 trillion package. Taiwan's public exposure is equity-market exposure plus a 6.38 percent shareholding: there is no Taiwanese policy-loan channel of comparable size in held data, and TSMC's 2025 accounts show a firm funding a US$40.9 billion capex year from operations (Observed, A/B). The demand side, US hyperscaler and datacenter debt, sets the size of the shock; it does not absorb the Korean supply-side loss, because the Korean firms' contracts are with accelerator buyers and foundries, not with the debt vehicles. Where HBM long-term agreements share downside with buyers (Section 2), part of the memory loss shifts to buyers' income statements; the share is not published and is a named gap.

Table 9: Scenarios: episode rates applied to the 2025 base (Modelled)

	2018–19 style	2022–23 style
Korean semiconductor exports, 2025 base (US$ bn)	173.4	173.4
Export fall applied (%)	−25.9	−23.7
Export revenue loss (US$ bn)	44.9	41.1
Samsung group operating profit (KRW tn)	20.6	6.6
SK hynix operating profit (KRW tn)	6.1	−53.6
TSMC revenue (US$ bn)	127.9	111.8
KOSPI earnings impact (%)	−29.5	−63.0
TAIEX earnings impact (%)	3.9	−7.5

Notes: Modelled (B/C). Firm rows apply the episode operating changes of Table 5 to 2025 operating profit (Samsung group KRW 43.6tn; SK hynix KRW 47.2tn) and the episode TSMC revenue change to 2025 revenue (US$122.4bn). The SK hynix 2022–23 cell applies a profit-to-loss swing rate, so the result is a loss of the scaled size. Index rows: KOSPI impact is the May 2026 two-firm weight (42.2 percent) times the mean of the two firms' episode operating changes. TAIEX impact is the TSMC weight (43 percent) times twice the TSMC revenue change, the factor of two standing for operating leverage at a 50.8 percent operating margin. These are arithmetic exposures, not forecasts. Source: results/scenario_korea_exports.csv; results/scenario_firms.csv.

Table 10: Financing map: quantified exposures (as announced, to November 2026)

Exposure	Amount	Bearer and channel
Korea support package (2024, raised 2025)	KRW 33tn	Korean state; fiscal and infrastructure support
of which low-interest policy loans	KRW 20tn	Korea Development Bank (state-owned); below-market rates to chip firms
SK hynix approved fabs Y2 and M17	KRW 54.3tn	SK hynix balance sheet, inside a KRW 600tn Yongin framework
Samsung domestic plan, 5 years	KRW 450tn	Samsung group balance sheet
TSMC capital budget 2026	US$60–64bn	TSMC balance sheet; 2025 net income US$55.2bn and operating cash flow above capex
TSMC US investment programme	US$165bn	TSMC, with US federal and state support not quantified in held data
Taiwan state shareholding in TSMC	6.38%	National Development Fund; equity exposure, no lending channel
Hynix creditor restructuring, 2001	not in held data	Creditor banks led by state lenders; the historical precedent for public absorption
US hyperscaler and datacenter debt	named gap	Demand side; no audited aggregate in held data (Section 4)

Notes: Amounts as announced (A/B). The table reports channels, not drawn balances: the KRW 20tn is a loan programme ceiling. Source: data/announcements.csv and Section 2 sources.

7 Limitations, sensitivity, and the falsification scorecard

Five limitations bind. (1) The DRAM fit ends in 2007 (named gap 4); if the post-2007 curve flattened, the treadmill arithmetic overstates current price pressure. The episode ASP falls of 2019 and 2023, which are post-2007 observations, are the check: both sit inside the fitted-curve range of outcomes. (2) The foundry proxy is blended revenue per wafer on a partial cumulative sum, grade C; a per-node fit could find learning inside nodes. MOPS and company disclosures do not publish a per-node price series, so the test cannot be run on public data. (3) Episode firm values for 2000–01 and 2007–09 are approximate (B/C); the scenario block uses only the two modern episodes, which are computed from the firm panel. (4) The break-even block is a representative-fab calculation with assumed cost shares (C); the grid, not the base case, carries the conclusion. (5) Market-cap concentration points are dated observations from exchange data via press reporting, not a continuous official series; the June 2026 monthly-average point is on a different basis and is flagged in Table 4.

Falsification scorecard, against the tests in Section 1. Test 1 (contract decoupling): partly met already. HBM is sold on annual contracts and 2026 reporting records downside-sharing long-term agreements (The Economist 2026); HBM was about a quarter to a third of Korean memory revenue on industry estimates, not enough to move the aggregate export response in 2023, when HBM was already material and chip exports still fell 23.7 percent. The thesis narrows accordingly: it is a claim about the commodity share, and it weakens as the contracted share rises. Test 2 (concentration falling): not met; every dated point since January 2025 rises. Test 3 (a 40 percent price fall without macro transmission): not observed in four episodes; the smallest export response to a fall of 43 percent or more was 19 percent.

Sensitivity. The headline arithmetic survives the weakest inputs: using the early-sample learning rate (35.1 percent) instead of the full-sample rate lowers the required volume growth at 20 percent bit growth from 13.7 to about 12 percent, and using the lowest fixed-cost cell raises the break-even fall only to 43.0 percent, the smallest episode fall on record.

8 Conclusion

The computations support a narrower thesis than the one commissioned. Reinvesting boom surpluses into capacity amplifies downturn damage where the product is commodity memory: the learning curve is steep (38.6 percent, steepening to 47.4 percent), the export and index concentration is at record measured levels, four episodes show export falls of 19 to 28 percent and profit swings that reach losses, and new-fab break-even price falls of 21 to 43 percent sit at or below every observed episode price fall. The thesis does not extend to leading-edge foundry output on current data: TSMC's price per wafer rises with cumulative output, its revenue fell in one year of the last nine, and it funds its capacity from operations. Korea has also pre-positioned the public balance sheet inside the exposure through KRW 20 trillion of policy loans; Taiwan has not, because on the evidence held it does not need to. The exposure to watch is the contracted share of Korean memory revenue: if it rises past the point where a 2023-scale price fall moves chip exports by less than 10 percent, the thesis fails on its own test.

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registered plan.

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A Data appendix: series definitions and sources

Table 11: Series definitions and sources

Series	Definition	Source
DRAM price, cumulative output	Price per kbit (2005 US$) and cumulative shipments, annual	PCDB compilation, reused from KR-WP-2026-05 data files (B)
Samsung, SK hynix panel	Group and semiconductor-division operating profit; hynix revenue; KRW tn	Company results releases and annual reports (B)
TSMC panel	Revenue, operating margin, capital expenditure; US$ bn	TSMC results releases and annual reports (A/B)
Korean exports	Semiconductor and total goods exports; US$ bn	MOTIE/MOTIR annual releases via press reporting; AMRO 2026 (B)
Taiwanese exports	Total goods exports; US$ bn	Ministry of Finance via Central Bank of Taiwan reporting (B)
TSMC wafer series	Revenue (NT$ bn) and shipments (thousand 12-inch equivalent wafers)	TSMC annual reports (C, reconstructed ratio)
Concentration points	Two-firm KOSPI share; TSMC TAIEX weight	KRX via Seoul Economic Daily and AJU Press; Bloomberg via The Edge (B/C)
Episodes	Percentage changes over each downturn window	Gartner; SIA/WSTS; company results; KRX/TWSE calendar returns via secondary compilations (B/C)
Announcements	Capacity, support, and trade amounts as announced	Company and government announcements (A/B)

Notes: Replication files: data/*.csv (inputs), results/*.csv (computed outputs), scripts/build_data.py, scripts/analysis.py, scripts/analysis2.py.